TAYLOR E. FIRMAN

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Experience

FRED HUTCHINSON CANCER CENTER, Seattle, WA 01/2024 – Present

Research Informatics Lead, Office of the Chief Data Officer

– Develop and maintain reproducible bioinformatic workflows using WDL and Docker containers, developing 6 genomic analysis pipelines and a library of 20 containerized software environments.
– Provide strategic guidance for infrastructure projects spanning on-premises and cloud computing resources, with focus on genomics and large-scale data analysis processes for translational research.
– Serve as Research Computing subject matter expert in Data House Calls advisory program, consulting with over 20 research teams on effective computing strategies across local cluster and cloud resources.
– Oversee deployment and maintenance of key research platforms including a local instance of cBioPortal, while developing templates and workflows to support genomic data management and analysis.

CHILDREN'S HOSPITAL COLORADO, Aurora, CO 06/2019 – 01/2024

Senior Bioinformatics Software Engineer, Precision Medicine Institute

– Develop bioinformatics pipelines to analyze samples for Oncology and Whole Exome Next Gen Sequencing panels. Weekly analysis runs examine over 2TB of data to identify relevant genetic variants and solidify patient diagnoses.
– Create, integrate, and maintain the Genomics Integration Engine (GENIE) to ensure seamless panel workflows from Epic ordering to analysis and result reporting, processing 8,500 panel orders serving more than 750 providers and 6,000 patients.
– Curate CHCO's Precision Diagnostics Database and develop API tools to enable cross-team database access, ensuring data integrity and supporting research initiatives.

CENTURYLINK, Broomfield, CO 05/2018 – 06/2019

Senior Data Scientist

– Perform audit of quote assessment tools to ensure acceptable financial standards, identifying over $2M in unprofitable deals.
– Develop centralized database for quote assessments and model business operations including revenue recognition standards and service optimization.

UNIVERSITY OF DENVER, Denver, CO 09/2012 – 05/2018

Graduate Teaching & Research Assistant

– Develop predictive computational models of biological systems across multiple scales, yielding novel analysis tools and insights published in peer-reviewed journals.
– Generate bioinformatic tools to mine protein and genome repositories, creating and organizing large-scale time-series databases for statistical analysis.
– Present research at ten scientific conferences and advance expertise in Python, SQL, and high-performance computing.

Education

THE UNIVERSITY OF DENVER, Denver, CO 08/2012 – 05/2018

Doctorate in Molecular and Cellular Biophysics, "Learning from disorder and noise in Physical Biology"

– 2017-18 Excellence in Research Award for the Division of Natural Sciences and Mathematics

THE UNIVERSITY OF PUGET SOUND, Tacoma, WA 09/2007 – 05/2011

Bachelor of Science in Physics and Mathematics

– Summa Cum Laude, Minor in French, CoSIDA/ESPN Academic All-American, NASA Research Scholar, Phi Beta Kappa

Peer Reviewed Publications

– **T. Firman**, J. Huihui, A.R. Clark, K. Ghosh (2021). Critical Comparison of MaxCal and Other Stochastic Modeling Approaches in Analysis of Gene Networks. *Entropy*.
– **T. Firman**, A. Amgalan, K. Ghosh (2018). Maximum Caliber Can Build and Infer Models of Oscillation in a Three-Gene Feedback Network. *Journal of Physical Chemistry B*.
– **T. Firman**, S. Wedekind, T.J. McMorrow, K. Ghosh (2018). Maximum Caliber can characterize genetic switches with multiple hidden species. *Journal of Physical Chemistry B*.
– **T. Firman**, K. Ghosh (2018). Sequence charge decoration dictates coil-globule transition in IDPs. *Journal of Chemical Physics*